Assignment 3 : Numerical Integration

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1. Assignment Statement

The objective of this assignment is to approximate the definite integral of the function (f(x)=e^{-x^2}) over the interval ([0,1]) using different numerical integration methods.

The integral to be calculated is:

$$I = \int_0^1 e^{-x^2} dx$$

The following methods are implemented in Python:

1. Calculate the integral using the exact reference value.
2. Apply the simple trapezoidal rule and calculate the absolute error.
3. Apply the composite trapezoidal rule with (n=4) and (n=6) subintervals, and calculate the errors.
4. Apply Simpson's rule and calculate the absolute error.
5. Apply the composite Simpson's rule with (n=4) and (n=6) subintervals, and calculate the errors.
6. Compare the results obtained using the composite trapezoidal and composite Simpson's methods.

2. Source Code

The Python implementation of this assignment is available on GitHub.

Repository link: https://github.com/HananeINHID/Fondements-mathematiques-IA.git

File: assignment3.py

3. The results

```
Reference value: 0.746824132812427

Simple trapezoid
Result : 0.6839397205857212
Error: 0.06288441222670582

 Composite trapezoid
n = 4
Result: 0.7429840978003812
Error: 0.0038400350120457727
n = 6
Result: 0.7451194124361793
Error: 0.0017047203762476926

Simple Simpson
Result: 0.7471804289095104
Error: 0.0003562960970834306

 Composite Simpson
n = 4
Result: 0.7468553797909873
Error: 3.124697856027314e-05
n = 6
Result: 0.7468303914893449
Error: 6.258676917947348e-06
```

4. Interpretation

4.1. Simple Trapezoidal Rule

The result obtained with the simple trapezoidal rule is 0.6839397205857212. The error is 0.06288441222670582, which is quite large compared to the other methods.

4.2. Composite Trapezoidal Rule

For $n = 4$, the result is 0.7429840978003812, with an error of 0.0038400350120457727.

For $n = 6$, the result is 0.7451194124361793, with an error of 0.0017047203762476926.

The result for $n = 6$ is closer to the reference value than the result for $n = 4$. This means that using more subintervals improves the result.

4.3. Simpson's Rule

Simpson's rule gives a result of 0.7471804289095104. The error is 0.0003562960970834306, which is much smaller than the error obtained with the simple trapezoidal rule.

4.4. Composite Simpson's Rule

For $n = 4$, the result is 0.7468553797909873, with an error of 0.00003124697856302714.

For $n = 6$, the result is 0.7468303914893449, with an error of 0.000006258676917947348.

The result for $n = 6$ is very close to the reference value. It also has a smaller error than the result for $n = 4$.

4.5. Comparison of Composite Methods

For both $n = 4$and $n = 6$, the composite Simpson's rule gives a result closer to the reference value than the composite trapezoidal rule. For example, with six subintervals, Simpson's rule has an error of only 0.00000626, compared to 0.00170472 for the trapezoidal rule.

Based on these results, I found that the composite Simpson's rule gives the best approximation for this integral.

5. Conclusion

In this exercise, We used different numerical methods to calculate the integral of e^{-x^2} between 0 and 1. The results show that some methods give better approximations than others. The simple trapezoidal rule has the largest error, while the results improve when we divide the interval into more subintervals.

Simpson's method gives better results than the trapezoidal method. The most accurate result is obtained with the composite Simpson's rule using $n = 6$, with an error of about 0.00000626.

In conclusion, the composite Simpson's rule with six subintervals gives the closest result to the reference value among the methods I tested.

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